Stuff We Already Know

- Given a right-angled triangle with angle x, write down the definitions of the following:

$$\sin x = _____, \cos x = _____$$

$$\tan x = _____, \sec x = _____$$

$$\csc x = _____, \cot x = _____$$

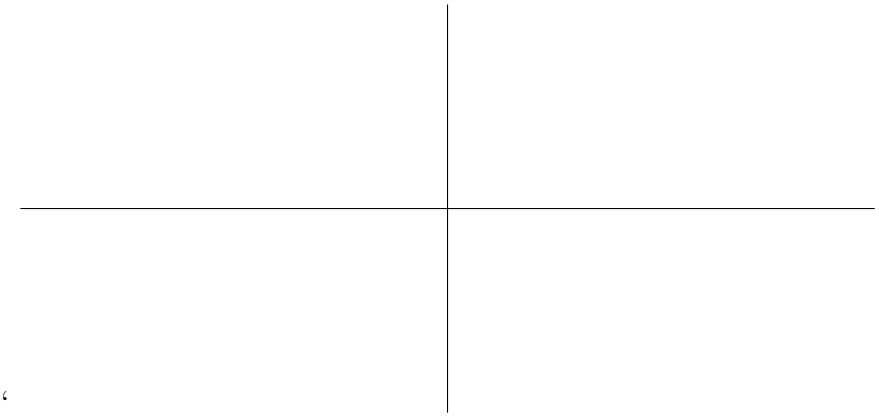
- Write down the following limits:

$$\lim_{x\to 0} \frac{\sin x}{x} = ____, \quad \lim_{x\to 0} \frac{\cos x - 1}{x} = ____.$$

- Write down the definition of the derivative for a function $f(x)$:

Derivatives of sin and cos

Use Geogebra to graph the function, $g(x) = \frac{\sin(x+0.001)-\sin(x)}{0.001}$ on the axes below with domain $[-2\pi, 2\pi]$. What does this approximate?



What other function that we know does this graph look like?

Write down a formula (using a limit) that would give the derivative of $\sin x$:

$$\frac{d}{dx}(\sin x) = \underline{\qquad\qquad\qquad\qquad}.$$

Use the fact that $\sin(A+B) = \sin A \cos B + \cos A \sin B$ to rewrite the derivative of $\sin(x)$ as the sum of two different limits.

Use your two limits to find what $\frac{d}{dx}\sin x$ is.

Question Use the same sorts of tricks to find $\frac{d}{dx}\cos x$:
(Note: $\cos(A+B) = \cos A \cos B - \sin A \sin B$.)

Other Trig Derivatives

Show that $\frac{d}{dx}\tan x = \sec^2 x$. (Hint: What's another way to write $\tan x$? Use the Quotient Rule.)

Show that $\frac{d}{dx}\sec x = \sec x \tan x$, $\frac{d}{dx}\csc x = -\csc x \cot x$, and $\frac{d}{dx}\cot x = -\csc^2 x$. (Hint: for example, write $\sec x = \frac{1}{\cos x} = (\cos x)^{-1}$ and use the chain rule...)

The derivatives we found above are important. While all trig functions have multiple forms (e.g. we write $\sec^2(x)$ instead of $\frac{1}{1-\sin^2(x)}$), these derivatives have standard forms. For future reference, write these below:

$\frac{d}{dx}\sin x =$ _____ $\frac{d}{dx}\cos x =$ _____

$\frac{d}{dx}\csc x =$ _____ $\frac{d}{dx}\sec x =$ _____

$\frac{d}{dx}\tan x =$ _____ $\frac{d}{dx}\cot x =$ _____

Questions

1. Find the equation of the tangent line to the graph of $f(x) = \tan x$ at $x = \frac{\pi}{4}$.

□Regrade

2. A particle is moving along a straight line. Its position from its origin (in meters) at time t seconds is given by $s(t) = \sin^2 t$. Find its velocity and acceleration at time $t = 2$ seconds. At what times is the particle at rest? (Hint: The identity $\sin 2x = 2 \sin x \cos x$ may be useful.)

□Regrade

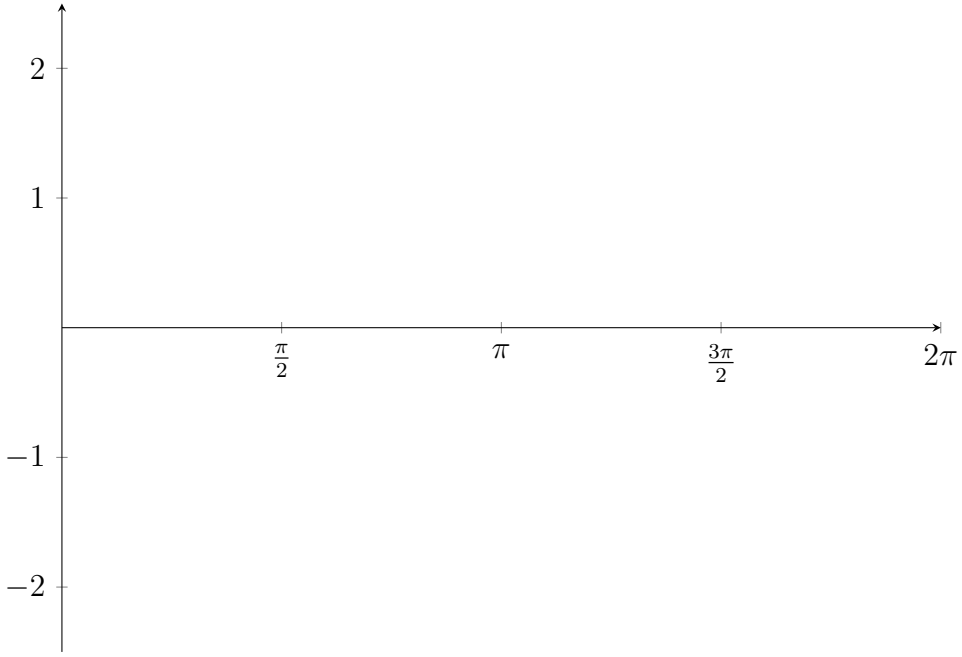
3. (a) Find the linear approximation to the curve $g(x) = \sec x$ at $x = \frac{5\pi}{6}$. Exact values with fractions, π, and roots here. No decimals.

□Regrade

(b) Use your answer above to estimate $\sec \frac{11\pi}{12}$.

4. □Regrade

(a) Use Geogebra to graph the function $y = \sin(2x) - 2\sin(x)$ over the horizontal range $[0, 2\pi]$. Insert special points on your graph to find decimal values of the points where the tangent line to the curve is horizontal [1]. Copy your graph and the x values of the points here.



(b) Use the derivative to find the exact x-coordinates of all points on the interval $[0, 2\pi]$ where the tangent line to the graph of the function $y = \sin(2x) - 2\sin(x)$ is horizontal. Check your exact answers approximate to the ones you got above.

(You will find the identity $\cos(2x) = 2\cos^2(x) - 1$ useful.)

[1] If you don't remember how to use Geogebra to do this, look back to the Derivatives and Roots lab from Math 105L!

5. Consider a particular point on a vibrating string as it moves vertically up and down. □Regrade The position of this point (in mm) at time t (in seconds) is given by

$$s(t) = 10 + \frac{1}{4}\sin(10\pi t).$$

(a) What is the period of oscillation of this point on the vibrating string?

(b) Find a formula for the velocity of the point on the string after t seconds.

(c) Describe the position and the motion (up or down) of this point on the string at $t = 0$ and $t = 0.3$ seconds. (Your answers should have units.)

6. If a projectile is fired from ground level with initial velocity v_0 and inclination angle □Regrade α and if air resistance can be ignored, the horizontal distance (in feet) it travels is

$$R = \frac{1}{16}v_0^2 \sin(\alpha)\cos(\alpha).$$

(a) Assuming that a soccer player kicks the ball at a 40° angle of inclination, find the initial velocity needed for a kick of 120 feet. (No calculus needed.)

(b) What value of α maximizes R?